TOPIC FOUR: MOMENT GENERATING FUNCTION (m.g.f)

4.0 Objectives

By the end of the topic, the leaner should be able to find the;

i) moment-generating function of the function of random variables

ii) use mgf to find the expectation and variance of random variables

4.1 Introduction

Computing all moments for a probability distribution function it's normally a tedious task

It is therefore desirable to have a function from which all the moments can be derived at will.

Such a function is called **a moment generating function**, m.g.f.

When this function exists for a particular distribution function, it is unique, a property which enables one to determine the probability distribution given the m.g.f

Definition

Let X be a random variable, the m.g.f denoted by $M_x(t)$ is defined as

$M_x(t) = E[e^{tx}]$ Where t is a small number, $-h < t < h$ and $h > 0$

The m.g.f is a special mathematical expectation of a random variable

If X is a discrete random variable with p.m.f$P(X = x)$ then its m.g.f $M_x(t) = E[e^{tx}]$

$= \sum_{all\ x} e^{tx}\ P(X = x)$ ….1

Similarly if X is a continuous random variable with p.d.f$f(x)$, then its m.g.f

$M_x(t) = E[e^{tx}] = \int_{-\infty}^{\infty} e^{tx}\ f(x)\ dx$ …2

The m.g.f in equation 1 and 2 exists if this summation is finite and also if the integral is also infinite

4.2 Derivation of Moments from M.G.F

We normally make use of the m.g.f to derive higher moments of a given random variable.

Let X be a discrete random variable, then its m.g.f is

$$M_x(t) = E(e^{tx}) = \sum_{all\ x} e^{tx}\, \Phi(X = x)$$

By the Taylors expansion e^{tx} can be expressed as

$$e^{tx} = 1 + tx + \frac{t^2x^2}{2!} + \frac{t^3x^3}{3} + \cdots$$

$$M_x(t) = \sum_{all\ x} (1 + tx + \frac{t^2x^2}{2!} + \frac{t^3x^3}{3} + \cdots)\, P(X = x)$$

$$= \sum_{all\ x} P(X = x) + \sum_{all\ x} tx\, P(X = x) + \sum_{all\ x} \frac{t^2x^2}{2!} P(X = x) + \sum_{all\ x} \frac{t^3x^3}{3} P(X = x) + \cdots$$

Differentiate both sides with respect to t to get

$$M_x^{'}(t) = 0 + \sum_{all\ x} x\, P(X = x) + \sum_{all\ x} tx^2\, P(X = x) + \sum_{all\ x} \frac{t^3x^3}{2}\, P(X = x) + \ldots$$

Set $t = 0$ in equation 1

$$M_x^{'}(0) = \sum_{all\ x} x\, P(X = x)$$

$$E(X) = u_1^{'}$$

Next differentiate equation 1 once more with respect to t to get

$$M_x^{''}(t) = 0 + \sum_{all\ x} x^2\, P(X = x) + \sum_{all\ x} \frac{tx^3}{1}\, P(X = x) + \cdots$$

Set $t = 0$

$$M_x^{''}(0) = \sum x^2 P(X = x)$$

$$= E(x^2) = u_2^{'}$$

In general, the rth raw moment $u_r^{'} = E(x^r) = M_x^r(t)$

The rth raw moment is derived by differentiating the m.g.f r times with respect to t and setting $t = 0$

These results can also be derived for a continuous random variable in analogous manner that is

$$M_x(t) = E[e^{tx}] = \int_{-\infty}^{\infty} e^{tx}\, f(x)dx$$

$$= \int_{-\infty}^{\infty} \left(1 + tx + + \frac{t^2x^2}{2!} + \frac{t^3x^3}{3} + \cdots\right) fx\, dx$$

$$= \int_{-\infty}^{\infty} f(x)\, dx + \int_{-\infty}^{\infty} tx\, f(x)\, dx + \int_{-\infty}^{\infty} \frac{t^2x^2}{2!} f(x)dx + \int_{-\infty}^{\infty} \frac{t^3x^3}{3!} f(x)dx + \cdots$$

$M_x^{'}(t) = \int_{-\infty}^{\infty} x\, f(x)dx + \int_{-\infty}^{\infty} \frac{tx^2}{1}\, f(x)dx + \int_{-\infty}^{\infty} \frac{t^2x^3}{2}\, f(x)dx + \cdots$ 1

Set $t = 0$

$$M_x^{'}(0) = \int_{-\infty}^{\infty} x\, f(x)\, dx = E(x)$$

Similarly if we find second derivative and set $t = 0$

$$M_x^{''}(0) = \int_{-\infty}^{\infty} x^2\, f(x)dx = E(x^2) = u_2^{'}$$

Therefore by definition to find mean and variance using mgf approach

$\boldsymbol{mean = E(X) = M_x^{'}(t)|_{set\ t=0} = M_x^{'}(0)}$

$\boldsymbol{variance = E\left(X^2 - \left(E(X)\right)^2\right) = (M_x^{''}(t) - \left(M_x^{'}(t)\right)^2\Big|_{set\ t=0} = M_x^{''}(0) - \left(M_x^{'}(0)\right)^2}$

Example

For each of the following random variables, find the mgf of

X is a discrete random variable, with pmf

$$P_x(X = k) = \begin{cases} \frac{1}{3} & k = 1 \\ \frac{2}{3} & k = 2 \end{cases}$$

Solution

$$M_x(t) = E(e^{xt}) = \frac{1}{3}e^t + \frac{2}{3}e^{2t}$$

SOME SPECIAL THEORETICAL PROBABILITY DISTRIBUTIONS (UNIVARIATE DISTRIBUTIONS)

These are theoretical distributions which are not contained by the actual experiments. They are deduced mathematically on the basis of certain assumptions

These distributions are broadly classified into 2 categories namely: discrete probability distributions and continuous distributions

TOPIC FIVE: DISCRETE DISTRIBUTIONS
5.0

Objectives

By the end of the topic, the learner should be able to;

i) Describe properties of uniform, Bernoulli, Binomial, Poisson and Geometric probability functions.
ii) Find expectation and variance of the random variables discrete distributions using direct method or the mgf approach
iii) Solve problems related to the discrete probability distibutions

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5.0 Objectives

By the end of the topic, the learner should be able to;

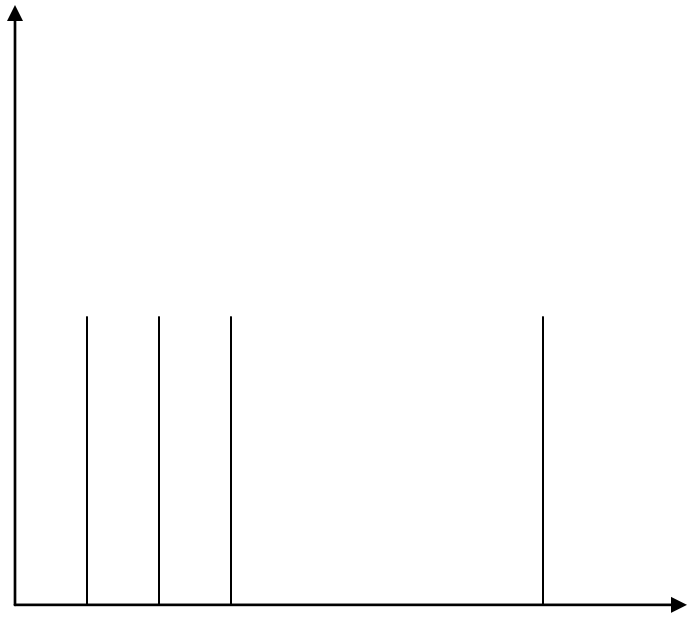
iv) Describe properties of probability functions of various discrete random variables

v) Find moments of the random variables

5.1 UNIFORM DISTRIBUTION

A Random variable X is defined to have a uniform distribution if the discrete density function of X is given by

$$f(x) = \frac{1}{N}, \qquad x = 1, 2, 3, \cdots, N$$
$$= 0, \quad elsewhere$$



The Mean, Variance and Moment generating function of the Uniform Distribution

$$Mean = E(x) = \sum_{x=1}^{N} x \cdot f(x)$$

$$= \sum_{x=1}^{N} x \cdot \frac{1}{N} = \frac{1}{N} \sum_{x=1}^{N} x$$

$$= \frac{1}{N}\left[1 + 2 + 3 + \cdots + N\right]$$

$$= \frac{1}{N}\left[\frac{N(N+1)}{2}\right] = \frac{N+1}{2}$$

$$\therefore \; Mean = \frac{N+1}{2}$$

$$Var(x) = \sigma^2 = E(x^2) - \left[E(x)\right]^2$$

Now we want to evaluate E (x^2).

$$E(x^2) = \sum_{x=1}^{N} x^2 \cdot f(x)$$

$$= \sum_{x=1}^{N} x^2 \cdot \frac{1}{N} = \frac{1}{N}\sum_{x=1}^{N} x^2$$

$$= \frac{1}{N}\left[1^2 + 2^2 + 3^2 + \cdots + N^2\right]$$

$$= \frac{1}{N}\left[\frac{N(N+1)(2N+1)}{6}\right] = \frac{(N+1)(2N+1)}{6}$$

$$\therefore\ E(x^2) = \frac{(N+1)(2N+1)}{6}$$

$$\therefore\ Var(x) = \sigma^2 = E(x^2) - \left[E(x)\right]^2$$

$$= \frac{(N+1)(2N+1)}{6} - \left[\frac{N+1}{2}\right]^2$$

$$= \frac{(N+1)(2N+1)}{6} - \frac{(N+1)^2}{4}$$

$$= \frac{N^2 - 1}{12}$$

$$\Rightarrow\ Var(x) = \frac{N^2 - 1}{12}$$

We want to evaluate the m.g.f.

$$M_X(t) = E(e^{tx})$$

$$= \sum_{x=1}^{N} e^{tx} \cdot f(x)$$

$$= \sum_{x=1}^{N} e^{tx} \cdot \frac{1}{N} = \frac{1}{N}\sum_{x=1}^{N} e^{tx}$$

$$= \frac{1}{N}\left[e^t + e^{2t} + e^{3t} + \cdots + e^{Nt}\right]$$

$$= \frac{e^t}{N}\left[1 + e^t + e^{2t} + \cdots + e^{(N-1)t}\right]$$

$$= \frac{e^t(1 - e^{Nt})}{N(1 - e^t)}$$

5.2 BERNOULLI DISTRIBUTION

Bernoulli trial is a random experiment with only two mutually exclusive outcomes that is success (occurrence of an event) or failure (non-occurrence of an event)

A good example of Bernoulli trial/e experiment is tossing a fair coin where we either obtain a head or a tail.

Let X be a random variable such that

$$x = \begin{cases} 1 \; if \; an \; outcome \; is \;\; a \; success \\ 0 \;\; if \; an \; outcome \; is \;\; a \; fail \end{cases}$$

This simple distribution is completely defined and is characterized by a single parameter P, where $P = \Pr(success)$ and $\Pr(X = 0) = 1 - p = q$

Therefore Bernoulli distribution can be expressed as $P(X = x) = \begin{cases} P^x(1-p)^{1-x}, x = 0,1 \\ 0, \; elsewhere \end{cases}$

Prove it is a pmf

$$\sum_{x=0}^{1} f(x) = \sum_{x=0}^{1} p^x \, q^{1-x}$$

$$= p^0 q^1 + p^1 q^0 = q + p = 1$$

$$\Rightarrow \;\; f(x) \;\; is \; a \;\; p.d.f.$$

The **moments** of this distribution are very easy to compute that is the rth raw moment

$$u_r^{'} = E[X^r] = \sum_{x=0}^{1} x^r P(X = x)$$

$$= \sum_{x=0}^{1} x^r p^x (1-p)^{1-x}$$

$$= 0 + p = p$$

$$u_1^{'} = p = E(X) = mean$$

Variance of X

$$var(x) = E(x^2) - \big(E(x)\big)^2$$

The m.g.f of X

$$M_x(t) = E[e^{tx}] = \sum_{x=0}^{1} e^{tx} \, p^x (1-p)^{1-x}$$

$$= \sum_{x=0}^{1} (e^t p)^x \, (1-p)^{1-x}$$

$$= 1 - p + e^t p$$

$$= q + pe^t$$

5.2 THE BINOMIAL DISTRIBUTION

We start with the binomial distribution, which is obtained if we are interested in the number of times an event A occurs in *n* independent performances of an experiment, assuming that A has a probability P (A) = p in a single trial. Then q = 1 – p is the probability that in a single trial the event A does not occur.

We assume that the experiment is performed *n* times and consider the random variable X = "number of times A occurs".

Then X can assume the values 0, 1, 2, …, n, and we want to determine the corresponding probabilities. For this purpose we consider any of these values, say, X = x, which means that in *x* of the *n* trials A occurs and in *(n – x)* trials it does not occur. This may look as follows:

AAA…A (x times) BBB…B ((n – x) times)…….(1)

Here B = A^c, i.e., A does not occur.

We assume that the trials are independent, i.e., do not influence each other.

Then, since P (A) = p and P (B) = q, we see that to Eq. (1) there corresponds the probability

ppp…p (x times) qqq…q ((n – x) times) = $p^x q^{n-x}$.

Clearly, Eq.(1) is merely **one** order of arranging x A's and (n – x) B's, and the probability P (X = x) thus equals $p^x q^{n-x}$ times the number of different arrangements of x A's and

(n – x) B's.

We may number the *n* trials from *1 to n* and pick *x* of these numbers corresponding to those trials in which A happens.

Since the order in which we pick the *x* numbers does not matter, we see that we can pick those *x* numbers from the *n* numbers in $\binom{n}{x}$ different ways.

Hence the probability corresponding to X = x equals

$$f(x) = \binom{n}{x} p^x q^{n-x}, \qquad x = 0,1,2,\cdots,n \qquad \text{.....(2)}$$
$$= 0, \ elsewhere$$

This is the probability that in *n*independent trials an event A occurs precisely *x* times where *p* is the probability of A in a single trial and *q = 1 – p*.

The distribution determined by the probability function (2) is called the **binomial distribution**.

The occurrence of A is called **success**, and the non-occurrence is called **failure**.

p is called the probability of success in a single trial.

Consider the function defined by

$$f(x) = \binom{n}{x} p^x q^{n-x}, \qquad x = 0,1,2,\cdots,n$$
$$= 0, \ elsewhere$$

where n is a positive integer and 0<p<1. *p* = probability of success and *q* is the probability of failure.

Under these conditions it is clear that $f(x) \geq 0$ and that

$$\sum_{x=0}^{n}\binom{n}{x}p^x q^{n-x} = \binom{n}{0}p^0 q^{n-0} + \binom{n}{1}p^1 q^{n-1} + \binom{n}{2}p^2 q^{n-2} + \cdots + \binom{n}{n}p^n q^0$$

$$= q^n + \binom{n}{1}pq^{n-1} + \binom{n}{2}p^2 q^{n-2} + \cdots + p^n$$

$$= (p+q)^n = 1$$

That is, $f(x)$ satisfies the conditions of being a p.d.f. of a random variable X of the discrete type.

A random variable X that has a p.d.f. of the form $f(x)$ is said to have a **binomial** distribution, and any such $f(x)$ is called a **binomial p.d.f.**

A binomial distribution will be denoted by the symbol b (n, p). The constants n and p are called the **parameters** of the binomial distribution.

Thus, if we say X is b (5, 1/3), we mean that X has the binomial p.d.f.

$$f(x) = \binom{5}{x}\left(\frac{1}{3}\right)^x\left(\frac{2}{3}\right)^{5-x}, \qquad x = 0,1,2,3,4,5$$

$$= 0, \ elsewhere$$

Remark

The binomial distribution serves as an excellent mathematical model in a number of experimental situations, e.g. a random experiment, the outcome of which can be classified in but one of the two mutually exclusive and exhaustive ways, say, success or failure (for example, head or tail, life or death, effective or noneffective, e.t.c.)

The Mean, Variance and Moment generating function of the Binomial Distribution

We first work out the m.g.f. which will in turn be used to obtain the mean and variance expressions.

$$M.g.f. = M_X(t) = E(e^{tx})$$

$$= \sum_{x=0}^{n} e^{tx} \cdot \binom{n}{x} p^x q^{n-x} = \sum_{x=0}^{n} \binom{n}{x} e^{tx} p^x q^{n-x} = \sum_{x=0}^{n} \binom{n}{x} \left(pe^t\right)^x q^{n-x}$$

$$= \binom{n}{0}\left(pe^t\right)^0 q^n + \binom{n}{1}\left(pe^t\right)^1 q^{n-1} + \binom{n}{2}\left(pe^t\right)^2 q^{n-2} + \binom{n}{3}\left(pe^t\right)^3 q^{n-3} + \cdots + \binom{n}{n}\left(pe^t\right)^n q^0$$

$$= q^n + \binom{n}{1}\left(pe^t\right)^1 q^{n-1} + \binom{n}{2}\left(pe^t\right)^2 q^{n-2} + \binom{n}{3}\left(pe^t\right)^3 q^{n-3} + \cdots + \left(pe^t\right)^n$$

But we know that from the Binomial formula of expansion

$$(a+b)^n = a^n + \binom{n}{1} a^{n-1}b^1 + \binom{n}{2} a^{n-2} b^2 + \cdots + b^n$$

In our case;

a = q and b = pe^t

$$\left(q + pe^t\right)^n = q^n + \binom{n}{1}\left(pe^t\right)^1 q^{n-1} + \binom{n}{2}\left(pe^t\right)^2 q^{n-2} + \binom{n}{3}\left(pe^t\right)^3 q^{n-3} + \cdots + \left(pe^t\right)^n$$

$$\Rightarrow M_X(t) = \left(q + pe^t\right)^n$$

Next, we know that

$$Mean = M_X'(0)$$

$$\therefore M_X'(t) = \frac{d}{dt}\left[\left(q + pe^t\right)^n\right]$$

$$= n\left(q + pe^t\right)^{n-1} pe^t$$

$$= npe^t\left(q + pe^t\right)^{n-1}$$

at t = 0; we have

$$M_X^{'}(0) = n\left(q + pe^0\right)^{n-1} pe^0$$
$$= np(p+q)^{n-1}$$
$$= np \quad Since \quad p+q=1$$
$$\Rightarrow \quad Mean = np$$

Next we work out Var (x)

We know that

$$Var(x) = \sigma^2 = E(x^2) - \left[E(x)\right]^2$$
$$= M_X^{''}(0) - \left[M_X^{'}(0)\right]^2$$

$$M_X^{'}(t) = npe^t\left(q + pe^t\right)^{n-1}$$

Using product rule to differentiate the above expression, we get

Let

$$U = npe^t \Rightarrow \frac{dU}{dt} = npe^t$$
$$V = (q + pe^t)^{n-1} \Rightarrow \frac{dV}{dt} = (n-1)(q + pe^t)^{n-2} pe^t$$

$$\therefore \quad M_X^{''}(t) = V\frac{dU}{dt} + U\frac{dV}{dt}$$
$$= (q + pe^t)^{n-1} \cdot npe^t + npe^t \cdot (n-1)(q + pe^t)^{n-2} pe^t$$
$$= npe^t(q + pe^t)^{n-2}\left[(q + pe^t)^1 + (n-1)pe^t\right]$$
$$= npe^t(q + pe^t)^{n-2}\left[q + pe^t + npe^t - pe^t\right]$$
$$= npe^t(q + pe^t)^{n-2}\left[q + npe^t\right]$$

At t = 0;

$$M_X^{''}(0) = npe^0(q + pe^0)^{n-2}\left[q + npe^0\right]$$
$$= np(p+q)^{n-2}\left[q + np\right]$$
$$= np(q + np) \quad Since \quad p + q = 1$$
$$= npq + n^2p^2$$

But

$$Var(x) = M_X^{''}(0) - \left[M_X^{'}(0)\right]^2$$
$$= npq + n^2p^2 - (np)^2$$
$$= npq + n^2p^2 - n^2p^2$$
$$= npq$$

SPECIAL CASE:

When n = 1, the binomial distribution becomes the Bernoulli's distribution with

$$f(x) = \binom{n}{x} p^x q^{n-x}, \qquad x = 0,1$$
$$= 0, \ elsewhere$$

$$\Leftrightarrow f(x) = p^x q^{1-x}, \qquad x = 0,1$$
$$= 0, \ elsewhere$$

Note that

$$f(x) = p^x q^{1-x}, \qquad x = 0,1 \text{ is a p.d.f.}$$
$$= 0, \ elsewhere$$

Proof

$$\sum_{x=0}^{1} f(x) = \sum_{x=0}^{1} p^x q^{1-x}$$
$$= p^0q^1 + p^1q^0 = q + p = 1$$

$$\Rightarrow \quad f(x) \ is \ a \ p.d.f.$$

Example 1

The binomial distribution with p.d.f.

$$f(x) = \binom{7}{x}\left(\frac{1}{2}\right)^{x}\left(\frac{1}{2}\right)^{7-x}, \qquad x = 0,1,2,3,4,5,6,7$$
$$= 0, \ \textit{elsewhere}$$

has the moment-generating function

$$M_X(t) = \left(\frac{1}{2} + \frac{1}{2}e^{t}\right)^{7},$$

has $\textit{Mean} = np = 7 \cdot \frac{1}{2} = \frac{7}{2}$ *and* $\textit{Variance} = npq = 7 \cdot \frac{1}{2} \cdot \frac{1}{2} = \frac{7}{4}$.

Furthermore, if X is the random variable with this distribution, we have

$$\Pr(0 \le X \le 1) = \sum_{x=0}^{1} f(x) = f(0) + f(1)$$
$$= \binom{7}{0}\left(\frac{1}{2}\right)^{0}\left(\frac{1}{2}\right)^{7} + \binom{7}{1}\left(\frac{1}{2}\right)^{1}\left(\frac{1}{2}\right)^{7-1}$$
$$= (1)(1)\left(\frac{1}{2}\right)^{7} + (7)(1)\left(\frac{1}{2}\right)^{7} = \frac{1}{128} + \frac{7}{128} = \frac{8}{128}$$

and

$$\Pr(X = 5) = f(5)$$
$$= \binom{7}{5}\left(\frac{1}{2}\right)^{5}\left(\frac{1}{2}\right)^{2} = \frac{21}{128}$$

Example 2

The probability that a marksman will hit a target is $\frac{5}{6}$. He fires 9 shots.

Calculate the probability that he will hit the target:

(i) At least 7 times

(ii) No more than 6 times

If he hits the target exactly 6 times, calculate the probability that the three misses are with 3 successive shots.

Solution

$$\Pr(\textit{hitting target}) = \Pr(\textit{Success}) = \frac{5}{6}$$

$$\Pr(\textit{not hitting target}) = \Pr(\textit{Failure}) = 1 - \frac{5}{6} = \frac{1}{6}$$

Taking the nine shots to be independent events, the probability distribution of X, "the number of targets hit" is $Bin\left(9, \frac{5}{6}\right)$, and so

$$\Pr(X = x) = \binom{9}{x}\left(\frac{5}{6}\right)^{x}\left(\frac{1}{6}\right)^{9-x}, \quad x = 0,1,2,3,4,5,6,7,8,9$$
$$= 0, \textit{ elsewhere}$$

(i)

$$\Pr(\textit{at least 7 targets hit}) = \Pr(X \geq 7) = \Pr(X = 7) + \Pr(X = 8) + \Pr(X = 9)$$
$$= \binom{9}{7}\left(\frac{5}{6}\right)^{7}\left(\frac{1}{6}\right)^{2} + \binom{9}{8}\left(\frac{5}{6}\right)^{8}\left(\frac{1}{6}\right)^{1} + \binom{9}{9}\left(\frac{5}{6}\right)^{9}\left(\frac{1}{6}\right)^{0}$$
$$= 36 \cdot \left(\frac{5}{6}\right)^{7}\left(\frac{1}{6}\right)^{2} + 9 \cdot \left(\frac{5}{6}\right)^{8}\left(\frac{1}{6}\right)^{1} + 1 \cdot \left(\frac{5}{6}\right)^{9} = 0.822$$

(ii)

$$\Pr(\textit{no more than 6 targets hit}) = \Pr(X \leq 6)$$
$$= \Pr(X = 0) + \Pr(X = 1) + \Pr(X = 2) + \cdots + \Pr(X = 6)$$
$$= 1 - \left[\Pr(X = 7) + \Pr(X = 8) + \Pr(X = 9)\right]$$
$$= 1 - \binom{9}{7}\left(\frac{5}{6}\right)^{7}\left(\frac{1}{6}\right)^{2} - \binom{9}{8}\left(\frac{5}{6}\right)^{8}\left(\frac{1}{6}\right)^{1} - \binom{9}{9}\left(\frac{5}{6}\right)^{9}\left(\frac{1}{6}\right)^{0}$$
$$= 1 - 0.822 = 0.178$$

(iii) The number of ways of hitting the target exactly six times is ${}^{9}C_{6} = 84$.

The three misses happening with three successive shots can occur in 7 ways, i.e., 123, 234, 345, 456, 567, 678, or 789.

$$\therefore \Pr(6 \textit{ targets and 3 misses with successive shots}) = \frac{7}{84} = \frac{1}{12}$$

Example 3

Four coins are tossed simultaneously. Find the probability function of the random variable X = "number of heads" and compute the probabilities of obtaining:

(i) No heads

(ii) Precisely one head

(iii) At least one head.

Solution

$$\Pr(\textit{obtaining a head}) = \Pr(\textit{Success}) = \frac{1}{2}$$

$$\Pr(\textit{not obtaining a head}) = \Pr(\textit{Failure}) = 1 - \frac{1}{2} = \frac{1}{2}$$

Taking the four tosses to be independent events, the probability distribution of X, "the number of heads" is $Bin\left(4, \frac{1}{2}\right)$, and so

$$\Pr(X = x) = \binom{4}{x}\left(\frac{1}{2}\right)^{x}\left(\frac{1}{2}\right)^{4-x}, \quad x = 0,1,2,3,4$$
$$= 0, \textit{ elsewhere}$$

(i) Probability of obtaining no head = $\Pr(X = 0) = \binom{4}{0}\left(\frac{1}{2}\right)^{0}\left(\frac{1}{2}\right)^{4} = \frac{1}{16} = 0.0625$

(ii) Probability of obtaining one head = $\Pr(X = 1) = \binom{4}{1}\left(\frac{1}{2}\right)^{1}\left(\frac{1}{2}\right)^{3} = \frac{4}{16} = 0.25$

(iii) Probability of obtaining at least one head =

$$\Pr(X \geq 1) = \Pr(X = 1) + \Pr(X = 2) + \Pr(X = 3) = 1 - \Pr(X = 0)$$
$$= 1 - \binom{4}{0}\left(\frac{1}{2}\right)^{0}\left(\frac{1}{2}\right)^{4} = 1 - \frac{1}{16} = 1 - 0.0625 = 0.9375$$

Example 4

It is expected that 10% of production from a continuous process will be defective. Find the probability that in a sample of 10 units, chosen at random exactly 2 will be defective and at least 2 will be defective

Solution

Let the random variable x= number of defective items since our interest is on picking a defective item

So $X \sim bin(n,p)$

That is $P(X=x) = \begin{cases} \binom{n}{x} p^x q^{n-x}, x = 0,1,2,3,\dots,n \\ 0 \ \ elsewhere \end{cases}$

Now we need $P(X=2)$ but $n=10, p=0.1, q=1-p=0.9, x=2$

So $P(X=2) = \binom{10}{2}(0.1)^2(0.9)^{10-2} = 0.1937$

We also need $\Pr(X \geq 2) = \Pr(x=2) + \Pr(x=3) + \cdots + \Pr(x=10)$

$$= 1 - [\Pr(x=0) + \Pr(x=1)] = 0.264$$

EXERCISE

1. If the moment-generating function of a random variable X is $\left(\frac{1}{3} + \frac{2}{3}e^t\right)^5$,

   Find $\Pr(X = 2 \ or \ 3)$. $\left[\frac{40}{81}\right]$

2. Let Y be the number of successes throughout *n* independent repetitions of a random experiment having probability of success $p = \frac{1}{4}$.

   Determine the smallest value of n so that $\Pr(1 \leq Y) \geq 0.70$. **[5]**

3. Let X be bin (2, p) and let Y be bin (4, p).

   If $\Pr(X \geq 1) = \frac{5}{9}$, *find* $\Pr(Y \geq 1)$. $\left[\frac{65}{81}\right]$

4. Previous experience indicates that, of the students entering upon a particular diploma course, 90% will successfully complete it. One year, 15 students commence the course. Calculate the probability that:

   (i) All 15 successfully complete the course; **[0.206]**

(ii) Only 1 student fails; **[0.343]**

(iii) No more than 2 students fail; **[0.816]**

(iv) At least 2 students fail. **[0.451]**

5. A company has ten telephone lines. At any instant the probability that any particular line is engaged is $\frac{1}{5}$.

State the expected number of free telephone lines. **[8]**

Calculate for any instant, the probability that

(i) All the lines are enganged, **[0.0000001]**

(ii) At least one line is free, **[1]**

(iii) Exactly two lines are free. **[0.000074]**

6. Suppose that it is known that 10% of the glasses made by a certain glass-blowing machine will be defective in some way. If we randomly select 10 glasses made by this machine, what is the probability that none of them are defective? How many would we expect to be defective? **[0.349, 1]**

7. A manufacturer of small parts ships his parts in lots of size 20 to his customers. Assume each part either is or is not defective and that the probability an individual part is defective is 0.05.

(a) What is the expected number of defectives per lot? **[1]**

(b) What is the probability that any particular lot contains no defectives? **[0.358]**

8. Assume you are a customer of the manufacturer mentioned in question (7) above and you receive 10 lots from him.

(a) What is the expected number of lot that have no defectives? **[3.58]**

(b) What is the probability that you have received no defectives in the 10 lots? **[0.000035]**

9. The probability that pen drawn at random from a box of pens is defective is 0.1. If a sample of 6 pens is taken find, find the probability that it will contain

a) Number of defective pens

b) 5 or 6 defective pens

c) More than 2 defective pens

d) Less than 3 defective pens

5.3 THE POISSON DISTRIBUTION

A discrete random variable X having a probability density function (p.d.f) of the form

$$\Pr(X = x) = f(x) = \frac{e^{-\lambda}\lambda^{x}}{x!}, \quad x = 0, 1, 2, 3, \cdots$$
$$= 0, \textit{ elsewhere}$$

is said to have a **Poisson distribution**.

We write X is Po (□).

□ (>0) is the**parameter of the distribution**.

Note: There is no upper limit to the value of x. This is usually determined by practical considerations.

The Poisson recurrence formula is

$$\Pr(X = x+1) = \frac{\lambda}{x+1}\Pr(X = x) \quad for \quad x = 0, 1, 2, 3, \cdots$$

This enables successive probabilities to be more easily calculated once the initial probability is known.

USES

There are two main uses of the Poisson distribution.

1. Estimation of probabilities of random events which have a small probability of occurrence.

Typical applications of this are telephone calls arriving at a switchboard, insurance claims, accident rates, flaws in manufactured material.

Usually the mean rate of occurrence per unit time will be given, although this may be scaled accordingly.

2. Approximation to a binomial distribution with the same mean, i.e., □ = np, and usually n > 50 and p < 0.1.

The Moment generating function, Mean and Variance of the Poisson distribution

We first work out the m.g.f. which will in turn be used to obtain the mean and variance expressions.

By definition,

$$M.g.f. = M_X(t) = E(e^{tx})$$

$$= \sum_{x=0}^{n} e^{tx} \cdot f(x) = \sum_{x=0}^{n} e^{tx} \cdot \frac{e^{-\lambda}\lambda^x}{x!} = e^{-\lambda} \sum_{x=0}^{n} \frac{e^{tx}\lambda^x}{x!} = e^{-\lambda} \sum_{x=0}^{n} \frac{(\lambda e^t)^x}{x!}$$

$$= e^{-\lambda}\left[\frac{(\lambda e^t)^0}{0!} + \frac{(\lambda e^t)^1}{1!} + \frac{(\lambda e^t)^2}{2!} + \frac{(\lambda e^t)^3}{3!} + \cdots + \frac{(\lambda e^t)^n}{n!} + \cdots\right]$$

$$= e^{-\lambda}\left[1 + \lambda e^t + \frac{(\lambda e^t)^2}{2!} + \frac{(\lambda e^t)^3}{3!} + \cdots + \frac{(\lambda e^t)^n}{n!} + \cdots\right]$$

$$= e^{-\lambda} \cdot e^{\lambda e^t} = e^{-\lambda + \lambda e^t} = e^{\lambda(e^t - 1)}$$

$$\Rightarrow M_X(t) = e^{\lambda(e^t - 1)}$$

Next, we know that

$$Mean = M_X^{'}(0)$$

$$\therefore\ M_X^{'}(t) = \frac{d}{dt}\left[e^{\lambda(e^t-1)}\right]$$
$$= e^{\lambda(e^t-1)} \cdot \lambda e^t$$
$$= \lambda e^t e^{\lambda(e^t-1)}$$

Soat t = 0; we have

$$M_X^{'}(0) = \lambda e^0 e^{\lambda(e^0-1)}$$
$$= \lambda(1)(1)$$
$$= \lambda$$
$$\Rightarrow\ Mean = \lambda$$

Note: λ is a constant greater than zero.

Next we work out Var (x)

We know that

$$Var(x) = \sigma^2 = E(x^2) - \left[E(x)\right]^2$$
$$= M_X^{''}(0) - \left[M_X^{'}(0)\right]^2$$

$$M_X^{'}(t) = \lambda e^t e^{\lambda(e^t-1)}$$

Using product rule to differentiate the above expression, we get

Let

$$U = \lambda e^t \Rightarrow \frac{dU}{dt} = \lambda e^t$$

$$V = e^{\lambda(e^t-1)} \Rightarrow \frac{dV}{dt} = \lambda e^t e^{\lambda(e^t-1)}$$

$$\therefore \; M_X^{''}(t) = V\frac{dU}{dt} + U\frac{dV}{dt}$$

$$= e^{\lambda(e^t-1)} \cdot \lambda e^t + \lambda e^t \cdot \lambda e^t e^{\lambda(e^t-1)}$$

$$= \lambda e^t e^{\lambda(e^t-1)}\left[1 + \lambda e^t\right]$$

At t = 0;

$$M_X^{''}(0) = \lambda e^0 e^{\lambda(e^0-1)}\left[1 + \lambda e^0\right]$$

$$= \lambda(1)(1)\left[1 + \lambda\right]$$

$$= \lambda + \lambda^2$$

But

$$Var(x) = M_X^{''}(0) - \left[M_X^{'}(0)\right]^2$$

$$= \lambda + \lambda^2 - (\lambda)^2$$

$$= \lambda + \lambda^2 - \lambda^2$$

$$= \lambda$$

Thus, for a Poisson distributed random variable $Mean, \; \mu = \lambda = Var(X) = \lambda$

Poisson or Binomial

Given a frequency distribution of the binomial/Poisson type, the decision on whether to fit a binomial or Poisson distribution is often made by considering the mean and variance of the distribution.

The closer these two are in value the more likely a Poisson distribution is applicable.

Additive Property

The additive property of the Poisson distribution:

If X is Po (x) and Y is Po (y), then X + Y is Po (x + y).

Example 1

Telephone calls arrive at a switchboard at the rate of 50 per hour.

Find the probabilities of 0, 1, or 2 calls arriving in any 5 minute period.

Solution

The average rate of call per 5 minute period $= \frac{50}{12} = 4.17 calls$.

If the random variable X is "the number of calls in any 5 minute period",

then X is Po (4.17).

$$\Pr(X = x) = f(x) = \frac{e^{-4.17}(4.17)^x}{x!}, \quad x = 0, 1, 2, 3, \cdots$$
$$= 0, \ elsewhere$$

$$\Pr(X = 0) = f(0) = \frac{e^{-4.17}(4.17)^0}{0!} = 0.02$$

$$\Pr(X = 1) = f(1) = \frac{e^{-4.17}(4.17)^1}{1!} = 0.06$$

$$\Pr(X = 2) = f(2) = \frac{e^{-4.17}(4.17)^2}{2!} = 0.13$$

Example 2

A college professor, based on his past experience, feels that there is a probability of 0.001 that he will be late to any given class and that being late or not for any class has no effect on whether or not he is late for any other class.

Then the number of times X that he will be late to his next 100 classes is a binomial random variable with parameters n = 100 and p = 0.001, and the exact probabilities that he is late exactly 0 times and exactly 1 time are, respectively,

$$\Pr(X = 0) = \binom{100}{0} (0.001)^0 (0.999)^{100} = 0.9048$$

$$\Pr(X = 1) = \binom{100}{1} (0.001)^1 (0.999)^{99} = 0.0906$$

Note that in this case n is fairly large and p is rather small so that we would expect the preceding probabilities to be well approximated by those for a Poisson random variable with $\mu = np = 100 \times 0.001 = 0.1$. The Poisson values are:

$$\Pr(X = 0) = f(0) = \frac{e^{-0.1}(0.1)^0}{0!} = 0.9048$$

$$\Pr(X = 1) = f(1) = \frac{e^{-0.1}(0.1)^1}{1!} = 0.0905$$

and the approximation is quite good.

Example 3

Telephone calls reach a switchboard independently and at random, external ones at a mean rate of 1 in any 5 minute period, and internal ones at a mean rate of 2 in any 5 minute period.

Calculate the probability that there will be more than 2 calls in any period of 2 minutes.

Solution

Let the random variable E be "the number of external calls per 2 minute period",

E is Po (0.4).

Let the random variable I be "the number of internal calls per 2 minute period",

I isPo (0.8).

Using the additive property of Poisson distributions,

E + I is Po (0.4+0.8), i.e., Po (1.2).

$$\begin{aligned}\Pr(E+I>2) &= 1-\Pr(E+I\le 2)\\ &= 1-\left[\Pr(E+I=0)+\Pr(E+I=1)+\Pr(E+I=2)\right]\\ &= 1-\Pr(E+I=0)-\Pr(E+I=1)-\Pr(E+I=2)\end{aligned}$$

$$\Pr(E+I=0)=\frac{e^{-1.2}(1.2)^0}{0!}=e^{-1.2}=0.301$$

$$\Pr(E+I=1)=\frac{e^{-1.2}(1.2)^1}{1!}=1.2\cdot e^{-1.2}=0.361$$

$$\Pr(E+I=2)=\frac{e^{-1.2}(1.2)^2}{2!}=0.271$$

$$\begin{aligned}\therefore\ \Pr(E+I>2) &= 1-0.301-0.361-0.217\\ &= 0.121\end{aligned}$$

Example 4

The number of calls per 10 minutes received at a telephone switch boar d follows a Poisson distribution with mean 0.6. Find the probability that;

No calls will be recorded in the first 10 minutes

More than 2 calls will be received in a period of 40 minutes

Solution

Let the random variable X = number of calls received in the first 10 minutes

$$X \sim P_o(m = 0.6)$$

That is $\Pr(X = x) = \begin{cases} \frac{0.6^x e^{-0.6}}{x!}, & x = 0,1,2,\ldots \\ 0 \ \ elsewhere \end{cases}$

We need $\Pr(X = 0)$

$$= \frac{0.6^0 e^{-0.6}}{0!} = e^{-0.6} = 0.55$$

Let the random variable X be the number of calls received per 40 minutes, now if

$X \sim P_o(m = 0.6)$

Then $X' \sim P_o(0.6 \times 4)$

$$\Pr(X' = x') = \begin{cases} \frac{2.4^x e^{-2.4}}{x!}, & x = 0,1,2,3,4\ldots \\ 0 \ \ elsewhere \end{cases}$$

We need $\Pr(X' > 2) = 1 - [\Pr(X' = 0) + \Pr(X' = 1) + \Pr(X' = 2)$

$$1 - \left[\frac{2.4^0 e^{-2.4}}{0!} + \frac{2.4^1 e^{-2.4}}{1!} + \frac{2.4^2 e^{-2.4}}{2!}\right] = 0.430$$

Example 5

A certain hospital usually admits 50 patients per day. On average 3 patients in 100 require rooms provided with special facilities. On a certain day it was found that there are 3 such rooms available. Assuming that 50 patients will be admitted find the probability that more than 3 patients will require such special rooms.

Since n is large and P is small, we use Poisson application to binomial

So $P = 0.30, n = 50$

So that $m = np = 50 \times 0.03 = 1.5$

Let the random variable X= number of patients who require rooms with special facilities

So $X \sim P_o(m = 1.5)$

$$\Pr(X = x) = \begin{cases} \dfrac{1.5^x e^{-1.5}}{x!}, & x = 0,1,2,\ldots \\ 0 \ \ elsewhere \end{cases}$$

We need $\Pr(X > 3)\ 1 - [\Pr(X = 0) + \Pr(X = 1) + \Pr(X = 2) + \Pr(X = 3)]$

$$= 1 - \left[\frac{1.5^0 e^{-1.5}}{0!} + \frac{1.5^1 e^{-1.5}}{1!} + \frac{1.5^2 e^{-1.5}}{2!} + \frac{1.5^2 e^{-1.5}}{2!}\right] = 0.0638$$

EXERCISE

1. Telephone calls arrive at a switchboard at random intervals at an average rate of 24 calls per hour.

   Find the probability of receiving:

   (a) No calls in 5 minutes; **[0.135]**

   (b) More than 4 calls in 5 minutes; **[0.0527]**

2. The number of bacteria in 1 ml of inoculum has a Poisson distribution with Mean = 2.0. If at least 3 bacteria are needed for a dose to be ineffective, find the probability that a dose of 1 ml will cause infection. **[0.143]**

3. Two types of flaw, A and B, may occur in a manufactured cloth. The numbers of flaws of type A and of type B occurring per metre length of cloth are independent random variables having Poisson distributions with means 0.5 and 1, respectively.

   (a) Find the probabilities that a length of 1 metre of cloth will have:

   (i) 2 or fewer flaws of type A, **[0.986]**

   (ii) No flaw of either type. **[0.223]**

   (b) Show that the probability of a length of 1 metre of the cloth containing 1 flaw only is exactly three times that of it containing 1 flaw of each type.

   (c) Removing a type A flaw from the cloth costs 8 pence and removing a type B flaw costs 2 pence. Find the mean and the standard deviation of the cost of removing flaws per 1 metre length of cloth. **[mean = 6, s.d.= 2.45]**

4. Suppose that in the production of 50-ohm radio resistors, non-defective items are those that have a resistance between 45 and 50 ohms and the probability of a resistor's being defective is 0.2%. The resistors are sold in lots of 100, with the guarantee that all resistors are non-defective.
   What is the probability that a given lot will violate this guarantee? **[0.181]**
5. It has been observed that packages of KBL's beer are removed from the shelf of a particular supermarket at a rate of 10 per hour during rush periods. What is the probability that at least 1 package is removed during the first 6 minutes of a rush period? **[0.6321]**
   What is the probability that at least 1 package is removed from the shelf during each of 3 consecutive, nonoverlapping 6 – minute intervals?**[0.2526]**
6. At a certain manufacturing plant, accidents have been occurring at the rate of 1 every 2 months. Assuming the accidents occur independently, what is the expected number of accidents per year? What is the standard deviation of the number of accidents per year? What is the probability that there will be no accidents in a given month? **[6, 2.45, 0.61]**
7. Assume the sales made by a used car salesman occur like events in a Poisson process with parameter $\mu = 1$ per week. What is the probability he makes exactly 3 sales in a 2-week period? At least 3? At most 3? **[0.1804, 0.3233, 0.8571]**

8. A factory packs bolts in boxes of 500. The probability that a bolt is defective is 0.002. Find the probability that a box contains 2 defective bolts
9. The mean number of bacteria per Ml in liquid is known to be 4. Assuming that the number of bacteria follows a Poisson distribution.
   a) Find the probability that
      (i) In 1 ml of liquid there will be no bacteria
      (ii) There will be 4 bacteria
   b) Find the probability that
      (i) In 3 ml of liquid there will be less than 2 bacteria

(ii) In ½ ml of liquid there will be more than 2 bacteria

SAMPLING WITH AND WITHOUT REPLACEMENT

5.4 HYPERGEOMETRIC DISTRIBUTION

The binomial distribution is important in sampling with replacement. Suppose that a box contains N things, e.g., screws, m of which are defective. If we want to draw a screw at random, the probability of obtaining a defective screw is $p = m/N$.
Hence in drawing a sample of n screws **with replacement**, the probability that precisely x of the n screws are defective is

$$f(x) = \binom{n}{x} p^x q^{n-x} = \binom{n}{x} \left(\frac{m}{N}\right)^x \left(1-\frac{m}{N}\right)^{n-x}, \qquad x = 0,1,2,\cdots,n$$
$$= 0, \ elsewhere$$

In sampling **without replacement** the probability is

$$f(x) = \frac{\binom{m}{x} \times \binom{N-m}{n-x}}{\binom{N}{n}}, \quad x = 0,1,2,\cdots,n \qquad \ldots\ldots\ldots\

(c) $\binom{N-m}{n-x}$ different ways of picking *(n – x)* non-defectives from *N – m.*

and each way in (b) combined with each way in (c) gives the total number of mutually exclusive ways of obtaining *x* defectives in *n* drawings without replacement.

Since (a) is the total number of outcomes and we draw at random, each such way has the probability $\frac{1}{\binom{N}{n}}$. From this, equation (1) follows.

The hypergeometric distribution has the mean

$\mu = \frac{nm}{N}$ and the variance $\sigma^2 = \frac{nm(N-m)(N-n)}{N^2(N-1)}$.

Proof (Exercise)

Example 1

A carton contains 20 fuses, 5 of which are defective. Find the probability that, if a sample of 3 fuses is chosen from the carton by random sampling without replacement, *x* fuses in the sample will be defective.

Solution

We have N = 20, m = 5, N – m = 15, n = 3.

For sampling without replacement the probability is

$$f(x) = \frac{\binom{m}{x} \times \binom{N-m}{n-x}}{\binom{N}{n}} = \frac{\binom{5}{x} \times \binom{15}{3-x}}{\binom{20}{3}}, \quad x = 0,1,2,3$$

$$f(0)=\frac{\binom{5}{0}\times\binom{15}{3}}{\binom{20}{3}}=\frac{91}{228}\ ,\ f(1)=\frac{\binom{5}{1}\times\binom{15}{2}}{\binom{20}{3}}=\frac{35}{76}\ ,\ f(2)=\frac{\binom{5}{2}\times\binom{15}{1}}{\binom{20}{3}}=\frac{5}{38}.$$

$$f(3)=\frac{\binom{5}{3}\times\binom{15}{0}}{\binom{20}{3}}=\frac{1}{114}$$

Example 2

A box of 20 spare parts for a given type of machine contains 15 good parts and 5 defective parts, if 4 parts are selected by chance from the box, what is the probability that exactly 3 of the parts are good

Solution

Let the random variable x = number of good parts

So X follows hyper geometric distribution

That is $P(X=x)=\begin{cases}\frac{\binom{k}{x}\binom{N-k}{n-x}}{\binom{N}{n}}, x=0,1,2,3,\dots\\ 0 \quad elsewhere\end{cases}$

$$N=20, n=4, k=15, x=3, n-x=1$$

We need $P(X=3)=\frac{\binom{15}{3}\binom{5}{1}}{\binom{20}{4}}=0.4696$

Example 3

Think of a basket which contains two types of balls, black and white, drawing a white ball as such if the variable N describes number of all the basket and the number of white balls, then $N-m=number\ of\ base\ balls$

Now assume there are 50 white and 45 blacks in the baskets close your eyes and draw 10 balls without replacement, what is the probability that 4 out 10 are white ones

Solution

$$N=50$$

$$m = 5$$

$$n = 10$$

$$x = 4$$

$$\Pr(X = 4) = \frac{\binom{5}{4}\binom{50-5}{10-4}}{\binom{50}{10}} = 0.0396$$

EXERCISE

1. A box contains 5 marbles of which 3 are chipped. Two marbles are chosen randomly without replacement from the box. What is the probability function for the number of chipped marbles in the sample?
2. A panel of 7 judges is to decide which of 2 final contestants in a beauty contest will be declared winner; a simple majority of the judges will determine the winner. Assume 4 of the judges will vote for Marie and that the other 3 will vote for Sue. If we randomly select 3 of the judges and ask them who they are going to vote for, what is the probability that a majority of the judges in the sample will favour Marie? **[22/35]**
3. An incoming lot of material contains 100 items. A sample of 5 items will be inspected; each item selected will be classified as defective or non-defective. If the sample contains 1 or fewer defectives it will be accepted, otherwise it will be rejected. If the lot contains 5 defectives, what is the probability that it will be accepted? If the lot contains 15 defectives, what is the probability that it will be rejected? **[0.981, 0.8391]**
4. In a school there are 20 students 6 are compulsive smoker and keep cigarettes in their lockers all the time. One day prefects decide to check at random on 10 lockers. What is the probability that they would find cigarettes in at least3 of the lockers

$$P(\geq 3), N = 20, n = 10\ , x = 3, \ldots, 10$$

5.5 GEOMETRIC (PASCAL) DISTRIBUTION

A random variable X is defined to have a geometric distribution, if the discrete probability density function (p.d.f) of X is given by

$$f(x) = \; p\,q^{x}, \qquad x = 0,1,2,\cdots$$
$$= 0, \; elsewhere$$

0<p<1. p = probability of success and q is the probability of failure.

This distribution occurs in real life, in situations where we have say a random variable X representing the number of trials required before the first success, then X has a geometric distribution.

The Mean, Variance and Moment generating function of the Binomial Distribution

We first work out the m.g.f. which will in turn be used to obtain the mean and variance expressions.

By definition;

$$M.g.f. = M_X(t) = E(e^{tx})$$
$$= \sum_{x=0}^{\infty} e^{tx} \cdot f(x) = \sum_{x=0}^{\infty} e^{tx} pq^{x} = p\sum_{x=0}^{\infty} e^{tx} q^{x}$$
$$= p\sum_{x=0}^{\infty} (qe^{t})^{x}$$
$$= p\left[(qe^{t})^{0} + (qe^{t})^{1} + (qe^{t})^{2} + (qe^{t})^{3} + \cdots\right]$$
$$= p\left[1 + (qe^{t})^{1} + (qe^{t})^{2} + (qe^{t})^{3} + \cdots\right]$$
$$= p\left(1 - qe^{t}\right)^{-1} = \frac{p}{1 - qe^{t}}$$

$$\Rightarrow \; M_X(t) = \; p\left(1 - qe^{t}\right)^{-1}$$

Next, we know that

$$Mean = M_X^{'}(0)$$

$$\therefore\ M_X^{'}(t) = \frac{d}{dt}\left[p\left(1-qe^t\right)^{-1}\right] = p\frac{d}{dt}\left[\left(1-qe^t\right)^{-1}\right]$$
$$= p(-1)\left(1-qe^t\right)^{-2}(-qe^t)$$
$$= pqe^t\left(1-qe^t\right)^{-2}$$

at t = 0; we have

$$M_X^{'}(0) = pqe^0\left(1-qe^0\right)^{-2}$$
$$= pq(1-q)^{-2}$$
$$= pqp^{-2} \quad Since \quad p+q=1 \Rightarrow p=1-q$$
$$= qp^{-1}$$
$$\Rightarrow\ Mean = \frac{q}{p}$$

Next we work out Var (x)

We know that

$$Var(x) = \sigma^2 = E(x^2) - \left[E(x)\right]^2$$
$$= M_X^{''}(0) - \left[M_X^{'}(0)\right]^2$$

$$M_X^{'}(t) = pqe^t\left(1-qe^t\right)^{-2}$$

Using product rule to differentiate the above expression, we get

Let

$$U = pqe^t \Rightarrow \frac{dU}{dt} = pqe^t$$

$$V = (1-qe^t)^{-2} \Rightarrow \frac{dV}{dt} = (-2)(1-qe^t)^{-3}(-qe^t) = 2qe^t(1-qe^t)^{-3}$$

$$\begin{aligned}\therefore \quad M_X^{''}(t) &= V\frac{dU}{dt} + U\frac{dV}{dt} \\ &= (1-qe^t)^{-2}\cdot pqe^t + pqe^t \cdot 2qe^t(1-qe^t)^{-3} \\ &= pqe^t(1-qe^t)^{-3}\left[(1-qe^t)^1 + 2qe^t\right] \\ &= pqe^t(1-qe^t)^{-3}\left[1-qe^t + 2qe^t\right] \\ &= pqe^t(1-qe^t)^{-3}\left[1+qe^t\right]\end{aligned}$$

At t = 0;

$$\begin{aligned}M_X^{''}(0) &= pqe^0(1-qe^0)^{-3}\left[1+qe^0\right] \\ &= pq(1)(1-q)^{-3}\left[1+q\right] \\ &= pqp^{-3}\left[1+q\right] \quad Since \quad p+q=1 \Rightarrow p=1-q \\ &= qp^{-2}\left[1+q\right]\end{aligned}$$

But

$$\begin{aligned}Var(x) &= M_X^{''}(0) - \left[M_X^{'}(0)\right]^2 \\ &= qp^{-2}\left[1+q\right] - (qp^{-1})^2 \\ &= qp^{-2} + q^2p^{-2} - q^2p^{-2} \\ &= qp^{-2} = \frac{q}{p^2}\end{aligned}$$

5.6 NEGATIVE BINOMIAL DISTRIBUTION

Consider a sequence of independent repetitions of a random experiment with constant probability p of success.

Let the random variable X denote the total number of failures in this sequence before the rth success; i.e., X + r is equal to the number of trials necessary to produce exactly r successes. Here r is a fixed positive integer.

To determine the probability density function (p.d.f) of X, let x be an element of $\{x:\ x = 0,1,\ 2,3,\cdots\}$.

Then, by the multiplication rule of probabilities, $\Pr(X = x) = f(x)$ is equal to the product of the probability

$$\binom{x+r-1}{r-1} p^{r-1} q^{x}$$

of obtaining exactly (r – 1) successes in the first x + r – 1 trials and the probability p of a success on the $(x + r)^{th}$ trial.

Thus the p.d.f. $f(x)$ of X is given by

$$f(x) = \binom{x+r-1}{r-1} p^{r-1} q^{x} \cdot p = \binom{x+r-1}{r-1} p^{r} q^{x},\ x = 0,1,2,3,\cdots$$
$$= 0,\ elsewhere$$

A distribution with a p.d.f. of the form $f(x)$ is called a **negative binomial distribution**.

If r = 1; then X has the p.d.f.

$$f(x) = p\,q^{x},\ x = 0,1,2,3,\cdots$$
$$= 0,\ elsewhere$$

In this special case, r = 1, we say that X has a geometric distribution.

The Mean, Variance and Moment generating function of the Negative Binomial Distribution

We first work out the m.g.f. which will in turn be used to obtain the mean and variance expressions.

By definition;

$$
\begin{aligned}
M.g.f. = M_X(t) &= E(e^{tx}) \\
&= \sum_{x=0}^{\infty} e^{tx} \cdot f(x) = \sum_{x=0}^{\infty} e^{tx} \binom{x+r-1}{r-1} p^r q^x, \; = p^r \sum_{x=0}^{\infty} e^{tx} \binom{x+r-1}{r-1} q^x \\
&= p^r \sum_{x=0}^{\infty} \binom{x+r-1}{r-1} (qe^t)^x \\
&= p^r \left[\binom{r-1}{r-1} (qe^t)^0 + \binom{r}{r-1} (qe^t)^1 + \binom{r+1}{r-1} (qe^t)^2 + \binom{r+2}{r-1} (qe^t)^3 + \cdots \right] \\
&= p^r \left[1 + r \cdot (qe^t)^1 + \frac{r(r+1)}{2!} (qe^t)^2 + \frac{r(r+1)(r+2)}{3!} (qe^t)^3 + \cdots \right]
\end{aligned}
$$

Note that from:

$$
\begin{aligned}
(1-x)^{-n} &= 1 + \frac{(-n)(-x)}{1!} + \frac{(-n)(-n-1)(-x)^2}{2!} + \frac{(-n)(-n-1)(-n-2)(-x)^3}{3!} + \cdots \\
&= 1 + \frac{nx}{1!} + \frac{n(n+1)x^2}{2!} + \frac{n(n+1)(n+2)x^3}{3!} + \cdots
\end{aligned}
$$

Then;

$$
1 + r \cdot (qe^t)^1 + \frac{r(r+1)}{2!} (qe^t)^2 + \frac{r(r+1)(r+2)}{3!} (qe^t)^3 + \cdots = \left[1 - qe^t \right]^{-r}
$$

$$
\textit{Hence; } M_X(t) = p^r \left[1 - qe^t \right]^{-r} = \left[\frac{p}{1 - qe^t} \right]^r
$$

Next, we know that

$$Mean = M_X^{'}(0)$$

$$\therefore\ M_X^{'}(t) = \frac{d}{dt}\left[p^r\left(1-qe^t\right)^{-r}\right] = p^r\ \frac{d}{dt}\left[\left(1-qe^t\right)^{-r}\right]$$
$$= p^r\ (-r)\left(1-qe^t\right)^{-r-1}(-qe^t)$$
$$= p^r rqe^t\left(1-qe^t\right)^{-r-1}$$

at t = 0; we have

$$M_X^{'}(0) = p^r rqe^0\left(1-qe^0\right)^{-r-1}$$
$$= p^r rq(1)(1-q)^{-r-1}$$
$$= p^r rq(p)^{-r-1} \quad Since \quad p+q=1 \Rightarrow p=1-q$$
$$= rqp^{-1}$$
$$\Rightarrow\ Mean = \frac{rq}{p}$$

Next we work out Var (x)

We know that

$$Var(x) = \sigma^2 = E(x^2) - \left[E(x)\right]^2$$
$$= M_X^{''}(0) - \left[M_X^{'}(0)\right]^2$$

$$M_X^{'}(t) = p^r rqe^t\left(1-qe^t\right)^{-r-1}$$

Using product rule to differentiate the above expression, we get

Let

$$U = p^r rqe^t \Rightarrow \frac{dU}{dt} = p^r rqe^t$$

$$V = (1-qe^t)^{-r-1} \Rightarrow \frac{dV}{dt} = (-r-1)(1-qe^t)^{-r-2}(-qe^t) = (r+1)qe^t(1-qe^t)^{-r-2}$$

$$\begin{aligned}
\therefore \; M_X^{''}(t) &= V\frac{dU}{dt} + U\frac{dV}{dt} \\
&= (1-qe^t)^{-r-1} \cdot p^r rqe^t + p^r rqe^t \cdot (r+1)qe^t(1-qe^t)^{-r-2} \\
&= p^r rqe^t(1-qe^t)^{-r-2}\left[(1-qe^t)^1 + (r+1)qe^t\right] \\
&= p^r rqe^t(1-qe^t)^{-r-2}\left[1-qe^t + (r+1)qe^t\right] \\
&= p^r rqe^t(1-qe^t)^{-r-2}\left[1-qe^t + rqe^t + qe^t\right] \\
&= p^r rqe^t(1-qe^t)^{-r-2}\left[1+rqe^t\right]
\end{aligned}$$

At t = 0;

$$\begin{aligned}
M_X^{''}(0) &= p^r rqe^0(1-qe^0)^{-r-2}\left[1+rqe^0\right] \\
&= p^r rq(1)(1-q)^{-r-2}\left[1+rq\right] \\
&= p^r rq(1)(p)^{-r-2}\left[1+rq\right] \;\; Since \;\; p+q=1 \Rightarrow p=1-q \\
&= rqp^{-2}\left[1+rq\right]
\end{aligned}$$

But

$$\begin{aligned}
Var(x) &= M_X^{''}(0) - \left[M_X^{'}(0)\right]^2 \\
&= rqp^{-2}\left[1+rq\right] - (rqp^{-1})^2 \\
&= rqp^{-2} + r^2q^2p^{-2} - r^2q^2p^{-2} \\
&= rqp^{-2} = \frac{rq}{p^2}
\end{aligned}$$

EXAMPLE

A market research agency that conducts interviews by telephone has found, from past experience, that there is a 0.40 probability that a call made between 2.30 and 5.30 P.M. will be answered. Assuming a Bernoulli process:

(a) What is the probability that an interviewer's tenth answer comes on his twentieth call?

(b) What is the expected number of calls necessary to obtain seven answers?

(c) What is the probability that an interviewer will receive his first answer on his third call?

Solution

Let success denote an answer to a call. Then p = 0.40, x = 20 and k = 10

(a) Prob(10th answer comes on 20th call) = $\binom{20-1}{10-1}(0.4)^{10}(0.6)^{10} = 0.05856$

(b) Expected number of calls for seven answers = $\frac{k}{p}$ where k = 7 and p = 0.4

$= \frac{7}{0.4} = 17.5$

(c) Prob(1st answer comes on 3rd call) = $\binom{3-1}{1-1}(0.4)^{2}(0.6)^{2} = 0.144$

5.2